

Autonomous Lane Keeping on Tata Punch EV

Vehicle dynamics, perception, and closed-loop lateral control

AUTONOMY	CONTROLS	VEHICLE DYNAMICS
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Project overview

A summer 2026 engineering program centered on a lane-keeping architecture for a Tata Punch EV. The work links lane perception, path representation, lateral-error estimation, steering control, safety supervision, and repeatable validation.

Engineering problem

A useful lane-keeping prototype must convert uncertain road observations into bounded steering commands while respecting vehicle dynamics and preserving a clear path to safe disengagement.

System architecture - Perception: identify lane boundaries or a drivable corridor from camera data. - State estimation: calculate lateral displacement, heading error, curvature, and confidence. - Control: compare geometric and model-based steering laws, with saturation and rate limits. - Safety: require manual override, confidence thresholds, speed limits, and fault logging.	Engineering focus - Bicycle-model reasoning for lateral motion and steering response. - Controller tuning across speed, curvature, sensor noise, and actuator delay. - Simulation-first validation before any controlled physical-vehicle experiment.
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Verification plan

- Track RMS and peak lateral error, heading error, steering rate, and disengagements.
- Use scripted straight, constant-radius, S-curve, dropout, and delay scenarios.
- Document assumptions and keep public-road deployment outside prototype scope.

Status and evidence boundary Active summer 2026 design brief; implementation details and measured results should be added only after logged tests.
